Lectures on Quantum Computing and Completely Positive Maps

Day 5: positive definiteness and the leading principal minors

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Abstract

This document is generated by Isabelle from the theory `Day05_Sylvester_Criterion` of the session `Quantum_Computing_by_VP_Day05`. It closes item D5-1 of the audit of the formalised lecture notes: a Hermitian matrix is positive definite iff all its leading principal minors are positive. The theorem is not reproved: it is imported from the AFP entry *Two Hermitian Results* and restated in the notions of the formalised notes. The session is kept apart from the main document because that entry builds on HOL-Analysis, HOL-Probability and QHLProver; proofs are folded.

1 Day 5: Sylvester's criterion (D5-1)

For an $n \times n$ matrix $P = (p_{ij})$ and $k = 1, \ldots, n$, the k-th leading principal submatrix is $P_k = (p_{ij})_{1 \le i,j \le k}$, the upper left $k \times k$ corner, and $\det P_k$ is the k-th leading principal minor. On Day 5 the notes give, without proof:

Theorem (notes) (Sylvester's criterion)**.** Let $P = P^* \in M_n$. Then $P > 0$ (that is, $\langle x|Px\rangle > 0$ for every $x \neq 0$) if and only if $\det P_k > 0$ for $k = 1, \ldots, n$.

For Hermitian P each $\det P_k$ is real, since P_k is again Hermitian. The criterion does not extend to $P \ge 0$: $P = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}$ has minors $0, 0 \ge 0$ but $\langle e_2|Pe_2\rangle = -1$. For $n = 2$ it reads: $\begin{pmatrix} a & b \\ \bar{b} & c \end{pmatrix} > 0$ iff $a > 0$ and $ac - |b|^2 > 0$.

The criterion is proved in the AFP entry *Two Hermitian Results* (*Sylvester-Criterion.sylvester-criterion*), with its own notions of Hermitian matrix (*Complex-Matrix.hermitian*, from QHLProver) and positive definiteness (*positive-definite*, with the quadratic form *QF* and the order on *complex*). It is not reproved here. The lemmas below show that these notions agree with *Day04-Hermitian-Matrices.hermitia* and *pd* of the formalised notes; the theorem is then restated in those terms. Names that both libraries define (*hermitian*, *inner-prod*) are written with their theory prefix. Formally P_k is *lps* P k, and k ranges over $0, \ldots, n$: the case $k = 0$ adds nothing, since a 0×0 determinant is 1.

lemma *hermitian-iff-Complex-Matrix-hermitian*: *Day04-Hermitian-Matrices.hermitian A* $\longleftrightarrow$ *Complex-Matrix.hermitia A*
⟨*proof*⟩

lemma *quadratic-form-eq-inner-prod*:
fixes *P* :: *complex Matrix.mat* **and** *x* :: *complex Matrix.vec*
assumes *P*: *P* ∈ *carrier-mat n n* **and** *x*: *dim-vec x* = *n*
shows *quadratic-form P x* = *Quantum.inner-prod x* (*P* ∗*v x*)
⟨*proof*⟩

lemma *pd-iff-positive-definite*:
fixes *P* :: *complex Matrix.mat*

assumes *P*: $P \in$ *carrier-mat n n*
shows *Day04-Hermitian-Matrices.pd P* $\longleftrightarrow$ *positive-definite P*
⟨*proof*⟩

Sylvester's criterion: a Hermitian $P \in M_n$ is positive definite iff $\det P_k > 0$ for every leading principal submatrix P_k, $k \leq n$ (*lps*; the order is that of *complex*, so each minor is real and positive).

theorem *pd-iff-leading-minors-pos*:
fixes *P* :: *complex Matrix.mat*
assumes *P*: $P \in$ *carrier-mat n n* **and** *h*: *Day04-Hermitian-Matrices.hermitian P*
shows *Day04-Hermitian-Matrices.pd P* $\longleftrightarrow$ $(\forall\, k \leq n.\ 0 <$ *Determinant.det* (*lps P k*))
⟨*proof*⟩

corollary *pd-iff-leading-minors-real-pos*:
fixes *P* :: *complex Matrix.mat*
assumes *P*: $P \in$ *carrier-mat n n* **and** *h*: *Day04-Hermitian-Matrices.hermitian P*
shows *Day04-Hermitian-Matrices.pd P* $\longleftrightarrow$
$(\forall\, k \leq n.$ *Im* (*Determinant.det* (*lps P k*)) $= 0 \wedge 0 <$ *Re* (*Determinant.det* (*lps P k*)))
⟨*proof*⟩